

IP Addressing Table

VLAN, SVI & HSRP

Device	Interface	IP Address	Subnet Mask	Purpose
DSW1	VLAN 10	192.168.10.101	255.255.255.0	VLAN 10 SVI
DSW1	VLAN 20	192.168.20.101	255.255.255.0	VLAN 20 SVI
DSW1	VLAN 30	192.168.30.101	255.255.255.0	VLAN 30 SVI
DSW2	VLAN 10	192.168.10.102	255.255.255.0	VLAN 10 SVI
DSW2	VLAN 20	192.168.20.102	255.255.255.0	VLAN 20 SVI
DSW2	VLAN 30	192.168.30.102	255.255.255.0	VLAN 30 SVI
HSRP	VLAN 10	192.168.10.100	255.255.255.0	Virtual Gateway
HSRP	VLAN 20	192.168.20.100	255.255.255.0	Virtual Gateway
HSRP	VLAN 30	192.168.30.100	255.255.255.0	Virtual Gateway

HSRP roles:

- DSW1 → Active
- DSW2 → Standby

End Devices — User Site

Device	VLAN	IP Address	Subnet Mask	Default Gateway
PC0	VLAN 10	192.168.10.1	255.255.255.0	192.168.10.100
PC1	VLAN 10	192.168.10.2	255.255.255.0	192.168.10.100
PC2	VLAN 20	192.168.20.1	255.255.255.0	192.168.20.100

Device	VLAN	IP Address	Subnet Mask	Default Gateway
PC3	VLAN 20	192.168.20.2	255.255.255.0	192.168.20.100
PC4	VLAN 30	192.168.30.1	255.255.255.0	192.168.30.100
PC5	VLAN 30	192.168.30.2	255.255.255.0	192.168.30.100

Distribution Switch ↔ Edge Router

Device	Interface/Link	IP Address	Subnet Mask	Purpose
DSW1	Routed Link	10.10.10.1	255.255.255.252	DSW1 ↔ R1
R1	Routed Link	10.10.10.2	255.255.255.252	DSW1 ↔ R1
DSW2	Routed Link	10.10.10.5	255.255.255.252	DSW2 ↔ R1
R1	Routed Link	10.10.10.6	255.255.255.252	DSW2 ↔ R1

WAN / ISP

Device	Interface/Link	IP Address	Subnet Mask	Purpose
R1	WAN	100.1.1.1	255.255.255.252	R1 ↔ ISP
ISP Router	WAN	100.1.1.2	255.255.255.252	R1 ↔ ISP
ISP Router	WAN	101.1.1.2	255.255.255.252	ISP ↔ R2
R2	WAN	101.1.1.1	255.255.255.252	ISP ↔ R2
ISP Router	Loopback	8.8.8.8	—	Simulated Internet

GRE Tunnel

Device	Interface	IP Address	Subnet Mask	Purpose
R1	Tunnel 1	172.16.1.1	255.255.255.252	GRE Tunnel
R2	Tunnel 1	172.16.1.2	255.255.255.252	GRE Tunnel

GRE Tunnel Network: 172.16.1.0/30

Server Site

Device	Interface	IP Address	Subnet Mask	Default Gateway
HTTP Server	NIC	10.1.1.10	255.255.255.0	10.1.1.1
TFTP Server	NIC	10.1.1.20	255.255.255.0	10.1.1.1
FTP Server	NIC	10.1.1.30	255.255.255.0	10.1.1.1